

EC-467 Mobile Application Development



BS (CE)-2021 LAB REPORT # 1

Submitted By:

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation / Coding	Observation/ Program Results	Total
Total	5	5	5	15
Obtained				

Lab Instructor.

Dr. Muhammad Bilal

Development Environment and Gradle

Introduction:

The goal of this course is to teach the skills necessary to develop Android based applications using the Android Studio Integrated Development Environment (IDE) and the Android Software Development Kit (SDK). Beginning with the basics, this course provides an outline of the steps necessary to set up an Android development and testing environment. An overview of Android Studio is included covering areas such as tool windows, the code editor and the Designer tool. An introduction to the architecture of Android is followed by an in-depth look at the design of Android applications and user interfaces using the Android Studio environment. More advanced topics such as database management, content providers and intents are also covered, as are touch screen handling, camera access. Lab manuals of this course also cover advanced features of Android Studio such as Gradle build configuration and the implementation of build variants to target multiple Android device types from a single project code base. Assuming you already have some Java programming experience, are ready to download Android Studio and the Android SDK, have access to a Windows, Mac or Linux system and ideas for some apps to develop, you are ready to get started.

Gradle:

Gradle is an open-source build automation system that builds upon the concepts of Apache Ant and Apache Maven and introduces a Groovy-based domain-specific language (DSL) instead of the XML form used by Apache Maven for declaring the project configuration. Maven is a build automation tool used primarily for Java projects. Maven addresses two aspects of building software: first, it describes how software is built and second, it describes its dependencies.

Tools:

Android Studio 1.1

Practice Tasks:

This section will provide information about the practice tasks, which are required to be performed in lab session. Design solutions of problems discussed in each task and place the solution code in a folder specified by your lab instructor.

Solution:

In this lab we just introduced the android studio programming from the

web application.

Steps:

- 1.Go the web, first and foremost, application (Download android studio in your google search bar).
- 2.Besides click the download android studio hedgehog.
- 3.Presently trust that the application will get downloaded
- 4.After this you can fire set up your android studio by following the given advances.
- 5.In the wake of setting up your android studio you can begin your essential assignments.

Shortcomings:

The main came from the rendition that gradle was not coordinating with the few recent variants of the android studio.

The subsequent inadequacy came from the working of the emulator.The emulator was not working accurately because of the SDK manager and the API device was also not working properly in the android studio.

Troubleshooting:

For the main weakness I choose the SDK manager and I choose the redesign and install the latest version of API level which made it work accurately.

Now the second weakness of the I choose the layout because I need to make the layout of the mobile app and also to design the simulation panel of the phone which I went to emulator settings and choose the latest version of emulator in the android studio.

After, all of these investigation, my android studio work very smoth and efficient and got set up and I could at long last make my venture and get everything to work.

Conclusion:

In this lab,we learned how to install and set up the android studio application.